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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
not required.
Inclusion and Exclusion Criteria
Participants with invalid data on the outcome measure or relevant covariates were excluded in this study .
Attrition
In total, 509,886 variants and 2,293 individuals were remained after QC.
Sex as a biological variable
Characteristic L1-L4 BMD Total BMD Femur total BMD Participants , n 4,070 4,056 4,054 Sex , male ( ) 1,965 (48.28) 1,954 (48.18) 1,962
Subject Demographics
Age: Methods Definition of bone mineral density in UK Biobank The UK Biobank recruited about 500,000 participants aged between 40 and 69 years and conducted prospective studies on them from 2006 to 2010. 25 Briefly, the data field of BMD has three UK Biobank
Randomization
not detected.
Blinding
not detected.
Power Analysis
not detected.
Replication
not required.
Cell Line Authentication

Authentication: After estimating the proportion of gut microbiota variation explained by genetic variation among individuals, associations between genetic variants and specific microbial traits (MTs) were identified by fitting linear, logistic, multinomial, and multivariate regressions assuming an additive genetic model.
Data Information
Availability: 25 GWAS summary data of gut microbiota A total of 114 gut microbiota-related traits used in this study were derived from a recent publicly available large-scale GWAS of human gut microbiota.

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Software and Algorithms			
Allele calling was performed using GenomeStudio v.	GenomeStudio		Suggestion: GenomeStudio (RRID:SCR_010973)(link)
Copy number variants were called with PennCNV v.1.0.4.	PennCNV		Suggestion: PennCNV (RRID:SCR_002518)(link)
SNP variation was linkage disequilibrium (LD)-pruned using PLINK 2.0 and the flag --indep-pairwise 50 5 0.45. 28 In total, 3,321 LD independent loci associated with 16 S gut microbiota species phenotypes were identified to achieve genome-wide significance in the FGFP.	PLINK		Suggestion: PLINK (RRID:SCR_001757)(link)
The kinship coefficients were estimated by KING software (University of Virginia, USA) to remove the genetically related subjects.	KING		Suggestion: KING (RRID:SCR_009251)(link)

SciScore is an automated tool that is designed to assist expert reviewers by finding and presenting formulaic information scattered throughout a paper in a standard, easy to digest format. ***SciScore is not a substitute for expert review.*** SciScore also checks for the presence and correctness of several unique identifiers, including RRIDs (research resource identifiers) in the manuscript, detects sentences that appear to be missing RRIDs, and can even suggest RRIDs under certain circumstances. **All RRID suggestions should be verified;** only the author can know whether the suggestions are correct.

For a full description of scored criteria and tips for improving your score, please see https://www.scicrunch.com/sciscorereport-faq